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ML_ADOPTION_MARKERS_DICT = {
    "ml_frameworks_mentioned": [], # List[str]
    "specific_models_architectures": [], # List[str]
    "compute_resources_mentioned": [], # List[str]
    "datasets_referenced": [], # List[str]
    "ml_libraries_tools": [], # List[str]
    "ml_adoption_level": "none", # "none" | "minimal" | "moderate" | "substantial" | "core"
    "ml_application_domain": "", # str
    "ml_method_primary": None, # Optional[str]
    "integration_with_traditional_methods": False # bool
}

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REPRODUCIBILITY_INDICATORS_DICT = {
    "code_availability_mentioned": False, # bool
    "code_repository_type": None, # Optional[str]
    "data_availability_mentioned": False, # bool
    "data_sharing_statement": None, # Optional[str]
    "methodology_detail_level": "minimal", # "minimal" | "moderate" | "detailed" |
"comprehensive"
    "hyperparameters_specified": False, # bool
    "computational_environment_described": False, # bool
    "preprocessing_steps_detailed": False, # bool
    "statistical_methods_described": False, # bool
    "replication_feasibility": "not_feasible" # "not_feasible" | "difficult" | "moderate" |
"straightforward"
}

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RESEARCH_OUTCOMES_DICT = {
    "mentions_clinical_trials": False, # bool
    "clinical_trial_identifiers": [], # List[str]
    "mentions_patents": False, # bool
    "patent_numbers": [], # List[str]
    "mentions_corrections": False, # bool
    "mentions_retractions": False, # bool
    "validation_type": [], # List[str]
    "real_world_application_mentioned": False, # bool
    "commercialization_mentioned": False, # bool
    "regulatory_approval_mentioned": False # bool
}

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IMPACT_INDICATORS_DICT = {
    "claims_novelty": False, # bool
    "claims_improvement_over_existing": False, # bool
}

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"quantitative_improvements_mentioned": False, # bool
"comparison_to_baseline": False, # bool
"mentions_policy_implications": False, # bool
"mentions_clinical_guidelines": False, # bool
"mentions_media_coverage": False, # bool
"collaboration_indicators": [], # List[str]
"funding_sources_mentioned": [], # List[str]
"potential_impact_scope": "narrow" # "narrow" | "moderate" | "broad" | "transformative"
}

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PAPER_EVALUATION_DICT = {
    "paper_id": "", # str
    "field": "", # str
    "publication_date": None, # Optional[str]
    "ml_adoption": {
        "ml_frameworks_mentioned": [],
        "specific_models_architectures": [],
        "compute_resources_mentioned": [],
        "datasets_referenced": [],
        "ml_libraries_tools": [],
        "ml_adoption_level": "none",
        "ml_application_domain": "",
        "ml_method_primary": None,
        "integration_with_traditional_methods": False
    },
    "reproducibility": {
        "code_availability_mentioned": False,
        "code_repository_type": None,
        "data_availability_mentioned": False,
        "data_sharing_statement": None,
        "methodology_detail_level": "minimal",
        "hyperparameters_specified": False,
        "computational_environment_described": False,
        "preprocessing_steps_detailed": False,
        "statistical_methods_described": False,
        "replication_feasibility": "not_feasible"
    },
    "research_outcomes": {
        "mentions_clinical_trials": False,
        "clinical_trial_identifiers": [],
        "mentions_patents": False,
        "patent_numbers": [],
        "mentions_corrections": False,
        "mentions_retractions": False,
    }
}

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    "validation_type": [],
    "real_world_application_mentioned": False,
    "commercialization_mentioned": False,
    "regulatory_approval_mentioned": False
},
"impact_indicators": {
    "claims_novelty": False,
    "claims_improvement_over_existing": False,
    "quantitative_improvements_mentioned": False,
    "comparison_to_baseline": False,
    "mentions_policy_implications": False,
    "mentions_clinical_guidelines": False,
    "mentions_media_coverage": False,
    "collaboration_indicators": [],
    "funding_sources_mentioned": [],
    "potential_impact_scope": "narrow"
},
"overall_assessment": "", # str
"confidence_score": "low", # "low" | "medium" | "high"
"notes": "" # str
}

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Complete example with actual data:

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EXAMPLE_EVALUATION = {
    "paper_id": "249058541",
    "field": "Biology",
    "publication_date": "2022",
    "ml_adoption": {
        "ml_frameworks_mentioned": [],
        "specific_models_architectures": [],
        "compute_resources_mentioned": [],
        "datasets_referenced": [],
        "ml_libraries_tools": ["GraphPad Prism", "MS Excel"],
        "ml_adoption_level": "none",
        "ml_application_domain": "immunology - glycosylation analysis",
        "ml_method_primary": None,
        "integration_with_traditional_methods": False
    },
    "reproducibility": {
        "code_availability_mentioned": False,
        "code_repository_type": None,
        "data_availability_mentioned": False,
        "data_sharing_statement": None,
        "methodology_detail_level": "detailed",
    }
}

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    "hyperparameters_specified": False,
    "computational_environment_described": False,
    "preprocessing_steps_detailed": True,
    "statistical_methods_described": True,
    "replication_feasibility": "moderate"
  },
  "research_outcomes": {
    "mentions_clinical_trials": False,
    "clinical_trial_identifiers": [],
    "mentions_patents": False,
    "patent_numbers": [],
    "mentions_corrections": False,
    "mentions_retractions": False,
    "validation_type": ["statistical validation", "functional assays"],
    "real_world_application_mentioned": True,
    "commercialization_mentioned": False,
    "regulatory_approval_mentioned": False
  },
  "impact_indicators": {
    "claims_novelty": True,
    "claims_improvement_over_existing": False,
    "quantitative_improvements_mentioned": True,
    "comparison_to_baseline": True,
    "mentions_policy_implications": False,
    "mentions_clinical_guidelines": False,
    "mentions_media_coverage": False,
    "collaboration_indicators": ["multi-institutional"],
    "funding_sources_mentioned": [],
    "potential_impact_scope": "moderate"
  },
  "overall_assessment": "Traditional biology research with detailed methodology but no ML adoption. Focuses on glycosylation patterns in rheumatoid arthritis.",
  "confidence_score": "high",
  "notes": "Clear experimental design with statistical validation. No computational/ML methods used."
}

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